

# Why $A \oplus B$ is a Biproduct

by notes on Freyd & Scedrov, Categories · Allegories

*§1.591 — finite products and coproducts coincide in an abelian category*

In any **bicartesian** category a map from a coproduct to a product is a  $2 \times 2$  matrix of hom-set elements. In an **abelian** category the canonical comparison map  $u : A + B \rightarrow A \times B$  becomes an **isomorphism**. Its inverse is **not** a matrix — it must be built from the addition on hom-sets,  $v = \pi_A \iota_A + \pi_B \iota_B$ . We construct  $v$ , verify  $uv = 1$  and  $vu = 1$ , and watch the whole thing degenerate in SET.

**Remark 0.1** (Conventions (Freyd)).

Coproduct injections  $\iota_A : A \rightarrow A + B$ ,  $\iota_B : B \rightarrow A + B$ ; product projections  $\pi_A : A \times B \rightarrow A$ ,  $\pi_B : A \times B \rightarrow B$ . The hom-set  $(A, B) = \{\text{morphisms } A \rightarrow B\}$ .

## §1. The $2 \times 2$ matrix

A morphism **out of a coproduct** is one map per summand; a morphism **into a product** is one map per factor. Stacking the two universal properties, a single morphism is determined by four free morphisms.

**Definition 1.1** (Matrix of a map  $\text{sum} \rightarrow \text{product}$ ).

In any bicartesian category a morphism  $f : A_1 + A_2 \rightarrow B_1 \times B_2$  is **uniquely** described by

$$f \text{ is } \begin{pmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{pmatrix}, \quad f_{ij} = \iota_i f \pi_j \in (A_i, B_j).$$

The four entries are **unconstrained** hom-set elements. This much holds in SET too.

## §2. The canonical comparison map

Specialise to  $A_1 = B_1 = A$  and  $A_2 = B_2 = B$ . The **canonical map**  $u : A + B \rightarrow A \times B$  is the one carrying the identity pattern:

$$u \text{ is } \begin{pmatrix} 1_A & 0 \\ 0 & 1_B \end{pmatrix},$$

that is, the four equations

$$\begin{aligned} \iota_A u \pi_A &= 1_A, & \iota_A u \pi_B &= 0, \\ \iota_B u \pi_A &= 0, & \iota_B u \pi_B &= 1_B. \end{aligned}$$

### △ Zero morphism

The off-diagonal **0** is the **zero morphism** — the unique map through the zero object,  $0_{A,B} : A \rightarrow 0 \rightarrow B$ .

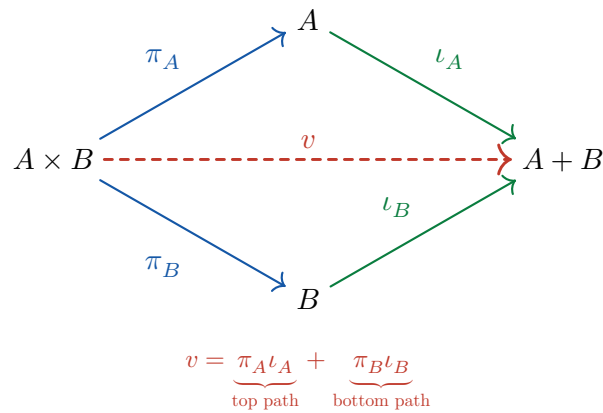
### §3. The inverse is not a matrix

The matrix trick described maps **sum**  $\rightarrow$  **product**. The inverse runs the other way,  $v : A \times B \rightarrow A + B$  — **out of a product, into a sum** — and **neither** direction has a componentwise universal property. So  $v$  cannot be read off a matrix; it must be **built**, using the **addition on hom-sets** that an abelian category provides.

**Definition 3.1** (The inverse, built with  $+$ ).

$$v := \pi_A \iota_A + \pi_B \iota_B \in (A \times B, A + B).$$

Here  $\pi_A \iota_A$  and  $\pi_B \iota_B$  are two morphisms  $A \times B \rightarrow A + B$ , summed in the abelian group  $(A \times B, A + B)$ . **This step is impossible in SET** — functions don't add.



Each summand is one corner trip — out along a projection  $\pi$  (blue), in along an injection  $\iota$  (green) — and the two trips are **added** (red) in the hom-group.

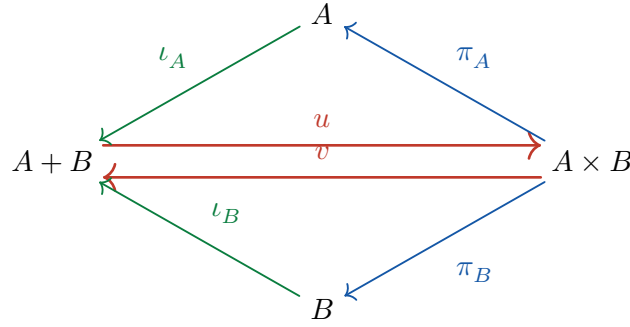
### §4. $u$ is an isomorphism

**Theorem 4.1** (Biproduct).

In an abelian category the canonical map  $u : A + B \rightarrow A \times B$  is an isomorphism, with inverse  $v = \pi_A \iota_A + \pi_B \iota_B$ . Hence finite products and coproducts coincide.

Both round trips collapse on **one** picture: the biproduct diamond carrying both  $u$  and  $v$ . Composition is bilinear, and the corner relations  $\iota_i u \pi_j = 1$  if  $i = j$ , else 0 (the matrix of  $u$ ) do all the work. We draw it twice — each time with the composite's **start** object on the left.

**Proof that  $uv = 1_{A+B}$**  — out of  $A + B$ , restrict along the injections  $\iota_A, \iota_B$ .

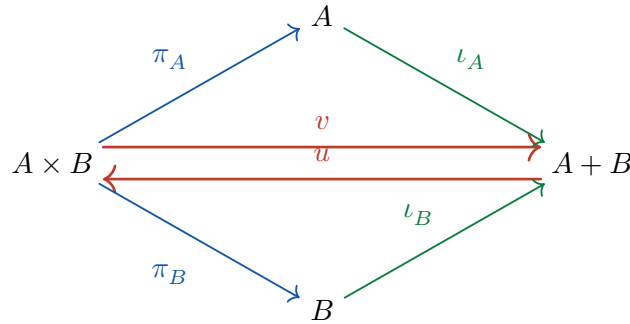


Going  $\iota_A \rightarrow u$  lands in  $A \times B$  with coordinates  $\iota_A u \pi_A = 1_A$  (top loop) and  $\iota_A u \pi_B = 0$  (crossing); then  $v = \pi_A \iota_A + \pi_B \iota_B$  reassembles them:

$$\iota_A u v = (\iota_A u \pi_A) \iota_A + (\iota_A u \pi_B) \iota_B = 1_A \iota_A + 0 = \iota_A,$$

and likewise  $\iota_B u v = \iota_B$ . A map out of  $A + B$  agreeing with 1 on both injections **is** 1.

**Proof that  $vu = 1_{A \times B}$**  — into  $A \times B$ , read coordinates along  $\pi_A, \pi_B$ . The same diamond, now  $A \times B$  on the left so the trip  $v \rightarrow u$  starts there:



Expanding  $v = \pi_A \iota_A + \pi_B \iota_B$  and using  $\iota_i u \pi_j \in \{1, 0\}$ :

$$vu \pi_A = \pi_A (\iota_A u \pi_A) + \pi_B (\iota_B u \pi_A) = \pi_A 1_A + \pi_B 0 = \pi_A,$$

and likewise  $vu \pi_B = \pi_B$ . A map into  $A \times B$  agreeing with 1 on both projections **is** 1. Hence  $u$  is an isomorphism.  $\square$

## §5. Concrete picture in $\mathbf{Ab}$

### Example.

In abelian groups both  $A + B$  and  $A \times B$  have the same underlying set  $A \oplus B = \{(a, b)\}$ , with

$$\iota_A : a \mapsto (a, 0_B), \quad \iota_B : b \mapsto (0_A, b), \quad \pi_A : (a, b) \mapsto a, \quad \pi_B : (a, b) \mapsto b.$$

The inverse **reassembles** a pair by adding its two one-sided pieces:

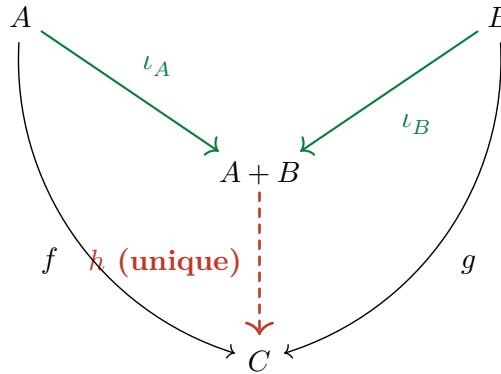
$$v(a, b) = \underbrace{(\pi_A \iota_A)(a, b)}_{\iota_A(a)} + \underbrace{(\pi_B \iota_B)(a, b)}_{\iota_B(b)} = (a, 0_B) + (0_A, b) = (a, b).$$

That single  $+$  is the **group operation** of  $A \oplus B$  — the abelian structure doing the work the universal properties cannot.

The map  $0 \rightarrow B$  sends the one element of the trivial group  $0 = \{0\}$  to  $B$ 's identity  $0_B$  (homomorphisms preserve identities), so the zero morphism  $A \rightarrow 0 \rightarrow B$  is the constant map  $a \mapsto 0_B$ .

## §6. Why the coproduct lands on the same set

The product  $A \times B = \{(a, b)\}$  is the usual Set product. The surprise is that the **coproduct** is the **same** set. By definition  $A + B$  is a coproduct iff for every pair of homomorphisms  $f : A \rightarrow C$ ,  $g : B \rightarrow C$  there is a **unique**  $h : A + B \rightarrow C$  with  $h\iota_A = f$  and  $h\iota_B = g$ :



**$h$  is forced.** Since  $(a, b) = \iota_A(a) + \iota_B(b) = (a, 0) + (0, b)$ , any such  $h$  obeys

$$h(a, b) = h\iota_A(a) + h\iota_B(b) = f(a) + g(b).$$

**$h$  is a homomorphism — but only because  $C$  is abelian.** Respecting  $+$  means

$$h((a, b) + (a', b')) = f(a + a') + g(b + b') = f(a) + f(a') + g(b) + g(b')$$

must equal  $h(a, b) + h(a', b') = f(a) + g(b) + f(a') + g(b')$  — and it does **only** after swapping  $g(b)$  past  $f(a')$ , i.e. using commutativity of  $C$ . In a non-abelian target this fails and the coproduct balloons into the free product  $A * B$ ; abelianness is exactly what collapses  $A + B$  back onto the product set  $\{(a, b)\}$ , with  $\iota_A(a) = (a, 0_B)$ ,  $\iota_B(b) = (0_A, b)$ .

### Theorem 6.1 (Punchline).

In SET you cannot add  $(a, 0_B)$  to  $(0_A, b)$ , there is no zero object, and

$$|A + B| = |A| + |B| \neq |A| \cdot |B| = |A \times B|.$$

Sums and products genuinely differ. In an abelian category the hom-sets are abelian groups, the inverse  $v = \pi_A \iota_A + \pi_B \iota_B$  exists, and  $A + B \cong A \times B$ : the **biproduct**  $A \oplus B$ .

## §7. Where the addition comes from: $+_L$ and $+_R$

The biproduct above was used as if hom-sets **already** carried a “+”. They do — and you do not have to assume it. The moment finite products and coproducts coincide (a **half-additive** category, zero object 0 included), every hom-set  $(A, B)$  acquires a canonical addition **for free**, built only from two structural maps the biproduct hands you:

**Definition 7.1** (Diagonal and codiagonal).

$$\Delta := \langle 1, 1 \rangle : A \rightarrow A \oplus A \quad (\text{duplicate; from the product side})$$

$$\nabla := [1, 1] : B \oplus B \rightarrow B \quad (\text{fold; from the coproduct side})$$

Given  $x, y \in (A, B)$  there are two obvious ways to splice them into a single map  $A \rightarrow B$ . In **diagram order** ( $xy$  = first  $x$  then  $y$ ):

**Definition 7.2** (The two additions).

$$x +_L y := A \xrightarrow{\Delta} A \oplus A \xrightarrow{[x, y]} B \quad (\text{duplicate, then copair})$$

$$x +_R y := A \xrightarrow{\langle x, y \rangle} B \oplus B \xrightarrow{\nabla} B \quad (\text{pair, then fold})$$

where  $[x, y]$  is the **coproduct** copairing and  $\langle x, y \rangle$  the **product** pairing.

The two are exact duals:  $+_L$  puts the work on the **target’s** coproduct side,  $+_R$  on the **source’s** product side. The whole content of §1.591 is that **they are the same operation** — and that this forces it to be commutative, associative, unital, and bilinear. That is the Eckmann–Hilton argument.

## §8. Element-wise, with $\exp$

The cleanest concrete instance uses **different** objects bridged by  $\exp$ , so the two operations involved are visibly distinct — addition on one side, multiplication on the other.

**Example** ( $A \neq B$  via  $\exp$ ).

Let  $A = (\mathbb{R}, +, 0)$  (reals, additive) and  $B = (\mathbb{R}_{>0}, *, 1)$  (positive reals, multiplicative). The continuous homomorphisms  $A \rightarrow B$  are the maps  $a \mapsto e^{ca}$ . Take

$$x : a \mapsto e^a, \quad y : a \mapsto e^{2a}.$$

Their sum should be the pointwise product in  $B$ :  $(x + y)(a) = e^a * e^{2a} = e^{3a}$ . Both recipes recover it.

**$+_L$  — copy in  $A$ , then copair into  $B$ .**

$$a \xrightarrow{\Delta} (a, a) \xrightarrow{[x, y]} x(a) * y(a) = e^a * e^{2a} = e^{3a}.$$

$\Delta$  merely **copies** (additive  $A$  contributes no operation); the copairing acts by  $[x, y](a_1, a_2) = x(a_1) * y(a_2)$ , the  $*$  being  $B$ ’s.

$+_R$  — pair into  $B \oplus B$ , then fold.

$$a \xrightarrow{\langle x, y \rangle} (e^a, e^{2a}) \xrightarrow{\nabla} e^a * e^{2a} = e^{3a}.$$

Here  $\nabla(b_1, b_2) = b_1 * b_2$  is the explicit last step.

Same hom  $a \mapsto e^{3a}$  either way. The only real operation is  $B$ 's  $*$ , firing once per route — inside  $[x, y]$  for  $+_L$ , inside  $\nabla$  for  $+_R$  — and exp makes the punchline literal:  $e^a * e^{2a} = e^{a+2a}$ , so multiplication in  $B$  realizes addition of “exponents”.

## §9. Why they must agree (Eckmann–Hilton)

**Theorem 9.1** ( $+_L = +_R$ , and it is a commutative monoid).

In any category with a zero object in which finite products and coproducts coincide,  $+_L$  and  $+_R$  define the **same** operation, which is associative, commutative, and has the zero morphism as unit; moreover composition distributes over it on both sides.

The engine is one identity — the **middle-two interchange** law, true because both sides are the same map read off the  $2 \times 2$  matrix  $\begin{pmatrix} u & v \\ x & y \end{pmatrix}$ :

$$(u +_R v) +_L (x +_R y) = (u +_L x) +_R (v +_L y).$$

Now specialise:

- set  $u = x = 0$ : it collapses to  $v +_R y = v +_L y$  — **the two operations are literally equal**. Drop the subscripts and write  $+$ .
- with subscripts gone the law is  $(u + v) + (x + y) = (u + x) + (v + y)$ ; set  $u = 0$  to get **associativity**, set  $u = y = 0$  to get **commutativity**.

So additivity is not extra data layered on top — it is **forced** by the biproduct, hence unique. This is exactly the rung Freyd needs: half-additive  $\rightarrow$  additive  $\rightarrow$  abelian.